AFFINE MORPHISMS: VANISHING, EXACTNESS AND DIRECT SUMS ON LQCOH-FBC

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Editorial note. Spot-check note: The algebraic-space-to-scheme reduction is delegated to 'the discussion in Cohomology of Spaces, Section spaces-cohomology-section-higher-direct-image' - a section-discussion reference rather than a numbered lemma.

Allowed standard prerequisites (editorial).

- Higher direct image restriction. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-loc-qcoh-fbc-compute-higher-direct-image. Reduction to algebraic spaces.
- Relative affine vanishing. Cohomology of Schemes, Lemma coherent-lemma-relative-affine-vanishing. Scheme-case vanishing.
- Affine equivalence of modules. Morphisms, Lemma morphisms-lemma-affine-equivalence-modules. Direct sums.

Source reference. Original proposition 0771: sources/stacks-chow/tags.txt, source lines 9186–9186. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GQH: sources/stacks-chow/tags.txt, source lines 20862–20862. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 071Y: sources/stacks-chow/tags.txt, source lines 9007–9007. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XC: sources/stacks-chow/tags.txt, source lines 2513–2513. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01SB: sources/stacks-chow/tags.txt, source lines 2332–2332. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be an affine morphism of algebraic stacks. The functor $f_* : LQCoh^{fbc}(\mathcal{O}_\mathcal{X}) \to LQCoh^{fbc}(\mathcal{O}_\mathcal{Y})$ is exact and commutes with direct sums. The functors $R^i f_*$ for $i > 0$ vanish on $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$.*

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Proof. The functors exist by Proposition 0771. By Lemma 0GQH this reduces to the case of an affine morphism of algebraic spaces taking higher direct images in the setting of quasi-coherent modules on algebraic spaces. By the discussion in Cohomology of Spaces, Section 071Y we reduce to the case of an affine morphism of schemes. For affine morphisms of schemes we have the vanishing of higher direct images on quasi-coherent modules by Cohomology of Schemes, Lemma 01XC. The vanishing for $R^1 f_*$ implies exactness of f_*. Commuting with direct sums follows from Morphisms, Lemma 01SB for example. □